

180-GBaud OOK Modulation Using a KNN Crystal-Based Electro-Optic Waveguide Modulator

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Abstract We demonstrate a KNN crystal-on-insulator Mach–Zehnder interferometer modulator operating in the C-band. The strong Pockels effect enables a low $V\pi \cdot L$ of 0.28 V·cm and 180-Gbaud OOK transmission, highlighting KNN as a promising platform for ultrahigh-speed, low-voltage optical modulation. ©2026 The Author(s)

Introduction

The rapid growth of bandwidth-intensive applications, including high-resolution video streaming, 6G communications, cloud services, and the Internet of Things, is driving a substantial increase in data traffic over optical-fiber networks. This trend requires energy-efficient and cost-effective optical transceivers and their constituent devices. Electro-optic (EO) waveguide modulators are particularly important because they can generate optical signals at symbol rates far exceeding 100 Gbaud while maintaining low drive voltages and power consumption.

Thin-film lithium niobate (TFLN) modulators have demonstrated optical signal transmission at symbol rates of up to 200 Gbaud [1,2]. However, the relatively modest EO coefficient of LN (~31 pm/V) limits further improvements in modulation efficiency and device footprint. Ferroelectric Pockels materials, including PZT, PLZT, and BTO, have therefore attracted attention as promising EO platforms [3]. Effective EO coefficients of 67, 120, and 342 pm/V have been reported for PZT-, PLZT-, and BTO-based devices, respectively [4–6]. Our previous measurements yielded half-wave voltage length products ($V\pi \cdot L$) of 1.3 V·cm for PZT, 1.1 V·cm for PLZT, and 0.60 V·cm for BTO modulators [7]. Although these modulators achieve low-voltage operation owing to their large EO coefficients, their relatively high permittivities can increase the device capacitance and microwave effective index, resulting in RF loss, impedance mismatch, and velocity mismatch between the optical and microwave signals.

In this work, we fabricated a potassium sodium niobate (KNN) crystalline film on a SiO₂/Si substrate and incorporated it into an EO waveguide modulator. The ferroelectric polarization in KNN can be aligned by applying a

DC electric field, enabling a large effective EO response. KNN has been reported to exhibit a moderate permittivity compared with conventional ferroelectric materials such as PZT [8]. This property is expected to mitigate excessive optical–microwave velocity mismatch in high-speed modulators. The fabricated MZI modulator exhibited a $V\pi$ of 0.70 V with a 4.0-mm-long phase shifter, corresponding to a $V\pi L$ of 0.28 V·cm, and successfully generated 180-Gbaud on–off keying signals.

Device Design and Fabrication

The deposition of a high-quality EO crystalline layer on an insulating substrate is critical for fabricating crystal-on-insulator modulators. The KNN films were prepared on 4-inch Si wafers covered with a 3-μm-thick thermally grown SiO₂ layer. Before KNN deposition, the SiO₂ surface was chemically modified with a thin intermediate layer to promote KNN crystallization with the desired orientation. KNN was subsequently deposited by sputtering at a substrate temperature of approximately 600 °C. The deposition process was continued until a film thickness of 300 nm was obtained.

The crystallographic properties of the KNN films were characterized by out-of-plane X-ray diffraction (XRD) using a θ – 2θ scan. Representative diffraction patterns on the substrate are shown in **Fig. 1(a)** and **1(b)**. The observed diffraction peaks confirm the formation of a well-crystallized KNN film with a preferred out-of-plane orientation. The chemically modified interface and thin intermediate layer promote the formation of the desired ferroelectric phase and crystal orientation on the SiO₂ surface. This controlled orientation is essential for obtaining a strong EO response in a coplanar-electrode modulator configuration. XRD and film-thickness measurements also

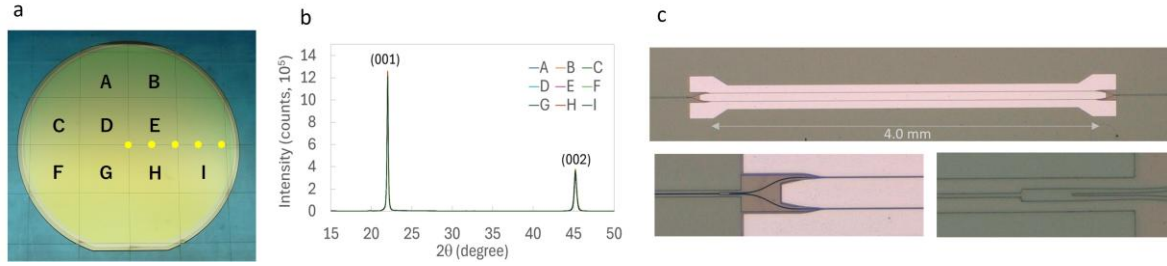


Fig. 1. KNN film and EO modulator. **a.** Photograph of a KNN film deposited on a 4-inch SiO_2/Si substrate. The thickness variation is less than 2% at the five measured points (yellow dots). **b.** XRD patterns of the 300-nm-thick KNN film measured at the locations labeled **A–I**, confirming its uniformity across the substrate. **c.** Top-view photograph of the KNN modulator (whole view, 1x2 part, and MMI part).

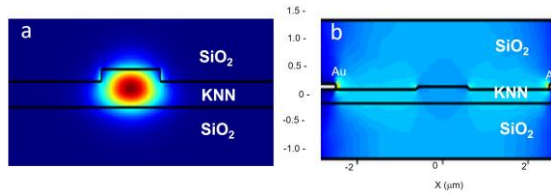


Fig. 2. **a.** Simulated optical electric-field distribution in the cross-section of the KNN waveguide. **b.** Simulated electric-field distribution.

confirmed good film uniformity across the wafer. In ferroelectric materials, electrically induced polarization is generally lost when the material is heated above its Curie temperature (T_c). KNN has a relatively high T_c of over 400 °C, providing favorable thermal stability during device fabrication and operation.

Device Structure and Characterization

Fig. 1(c) shows photo pictures of the fabricated MZI modulator. A ridge waveguide structure was employed to achieve single-mode operation. The simulated fundamental transverse-electric (TE_0) mode distribution is shown in **Fig. 2(a)**. The optical mode is strongly confined within the KNN layer because of the large refractive-index contrast between KNN (2.40) and SiO_2 . To support the TE_0 mode, the waveguide was designed with a ridge height of 100 nm and a ridge width of 1.6 μm . Multimode-interference (MMI) couplers were used as the optical splitter and combiner of the MZI.

CPW electrode was placed on both sides of the waveguide to apply a transverse electric field that interacts strongly with the TE-polarized optical mode. Simulated electric field distribution is shown in **Fig. 2(b)**. The effective Pockels response of the KNN modulator was enhanced by aligning the ferroelectric polarization under an applied DC electric field. Laser light at a wavelength of 1550 nm was coupled into the waveguide with TE polarization through a polarization-maintaining fiber. Low-frequency modulation characteristics were measured by applying a 20-kHz sinusoidal electrical signal to

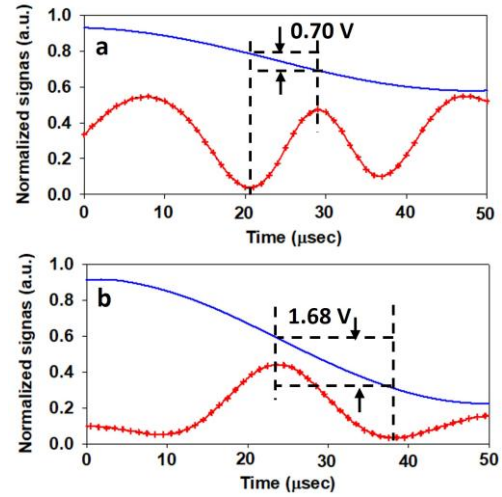


Fig. 3. Measurement of V_π . **a.** Modulation under a DC bias field of 15 $\text{V}/\mu\text{m}$. **b.** Bias-free modulation after poling the modulator with a DC electric field of 45 $\text{V}/\mu\text{m}$.

the electrodes. A bias tee was used to simultaneously apply a DC voltage for domain orientation in the KNN layer.

Figs. 3(a) and **3(b)** show the measurements of V_π under two different modulation conditions. Under an applied DC electric field of 15 $\text{V}/\mu\text{m}$, the measured V_π was 0.70 V for the 4.0-mm-long phase shifter, corresponding to a $V_\pi L$ product of 0.28 $\text{V}\cdot\text{cm}$. This enhanced modulation efficiency is attributed to DC-field-induced domain alignment and the resulting effective push–pull operation. After a higher poling field of 45 $\text{V}/\mu\text{m}$ was applied, remanent polarization was established in the KNN waveguide, eliminating the need for a DC bias during subsequent modulation. Under zero-DC-bias operation, a V_π of 1.68 V was measured for the 4.0-mm-long phase shifter, corresponding to a $V_\pi L$ of 0.67 $\text{V}\cdot\text{cm}$.

High-Frequency Modulation and Signaling

For modulation in the GHz regime, a traveling-wave electrode is required to efficiently deliver the RF drive signal along the phase shifter while mitigating the bandwidth limitation imposed by

the lumped device capacitance. Accordingly, the RF electrodes were designed to provide a characteristic impedance close to 50 Ω . The electrode structure comprises a coplanar transmission line along the phase shifter and ground–signal–ground (GSG) contact pads, enabling efficient RF coupling through GSG probes

For ferroelectric EO materials, their relatively high permittivity property must be carefully considered when designing modulators for high-frequency operation. In particular, the resulting microwave effective index should be considerably higher than the optical group index, making optical–microwave velocity matching a major challenge in achieving a broad modulation bandwidth. Compared with conventional ferroelectric-crystal platforms, the KNN-on-SiO₂ structure is expected to suppress an excessive increase in effective permittivity, thereby mitigating optical–microwave velocity mismatch.

We investigated the high-speed transmission capability of the KNN modulator by generating OOK signals at symbol rates of up to 180 Gbaud. A pseudorandom binary sequence (PRBS) generated by an 80-GHz arbitrary waveform generator (AWG; Keysight M8199B) was applied to the KNN modulator. A linear 100-GHz RF driver was used to set the drive-voltage swing to approximately 300 mV. The modulated optical output was amplified by an erbium-doped fiber amplifier (EDFA) and detected using a 100-GHz photodetector module (Fraunhofer HHI). The detected signals were analyzed using a high-speed digital communication analyzer (Keysight DCA-X N1000A with an N1046A module). A

poled KNN modulator was used in these high-speed signaling experiments.

The experimental setup is shown in **Fig. 4(a)**, and the measured eye diagrams at symbol rates of up to 180 Gbaud are presented in **Figs. 4(b)**. For 120, 160, and 180 Gbit/s OOK signals, corresponding bit-error ratios (BERs) are summarized. Following feed-forward equalization (FFE), the signal quality was improved, resulting in BERs below the soft-decision forward-error-correction (SD-FEC) threshold. Because the Pockels effect provides a linear relationship between the applied electric field and the induced refractive-index change, the KNN modulator is also expected to support higher-order modulation formats, including PAM4.

Conclusions

We demonstrated a high-efficiency traveling-wave MZM incorporating a KNN crystalline waveguide on a SiO₂/Si substrate. The modulator achieved a product of 0.28 V·cm under a DC field of 15 V/ μ m. After poling at 45 V/ μ m, remanent polarization enabled zero-DC-bias operation with a product of 0.68 V·cm. The relatively low permittivity of KNN is advantageous for reducing the microwave effective index and improving high-frequency performance. The device successfully generated 180-Gbaud OOK signals with a drive voltage of approximately 300 mV, achieving a BER below the SD-FEC threshold after FFE. These results establish KNN as a promising EO material platform for low-voltage, ultrahigh-speed optical modulators.

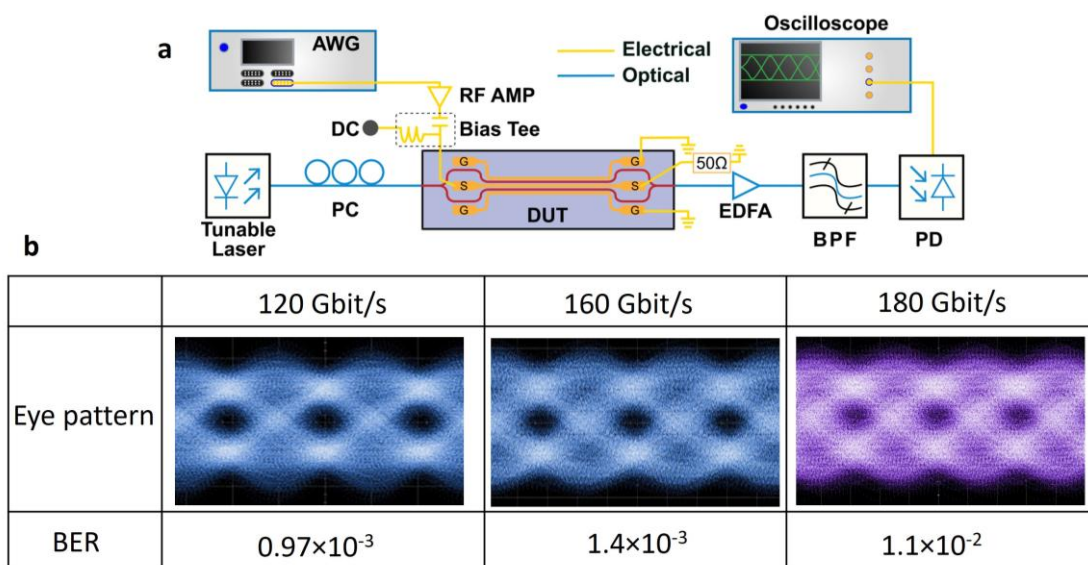


Fig. 4. High-speed OOK transmission experiment. **a.** Experimental setup used to generate and measure OOK signals from the KNN modulator. **b.** Eye diagrams obtained at data rates of 120, 160, and 180 Gbit/s. The estimated BER is indicated for each data rate.

Acknowledgements

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